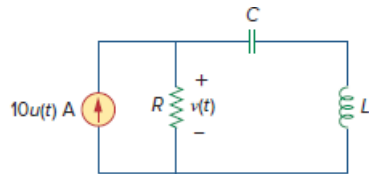


Problem

Find the response $v(t)$ for $t > 0$ in the circuit in Fig. 16.83. Let $R = 8\ \Omega$, $L = 2\text{ H}$, and $C = 125\text{ mF}$.

Figure 16.83:

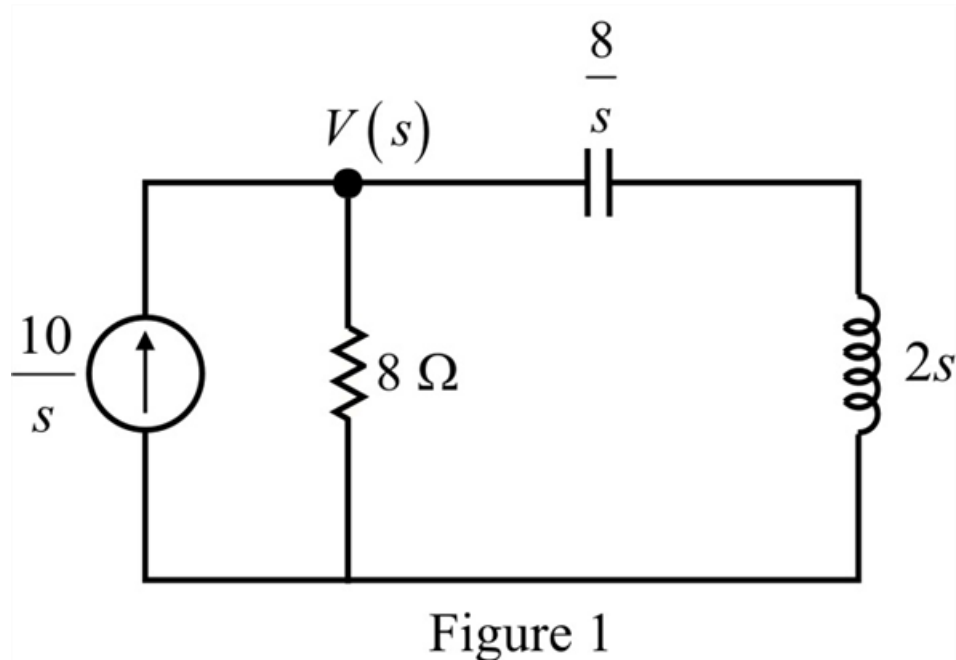


Step-by-step solution

Step 1 of 2

Refer to figure 16.83 for the circuit in the textbook.

Redraw the circuit in s-domain as shown in figure 1.



Step 2 of 2

Apply the Kirchhoff's current law at the node of voltage $V(s)$.

$$-\frac{10}{s} + \frac{V(s)}{8} + \frac{V(s)}{\frac{8}{s} + 2s} = 0$$

$$V(s) \left(\frac{1}{8} + \frac{1}{\frac{8}{s} + 2s} \right) = \frac{10}{s}$$

$$V(s) \left(\frac{1}{8} + \frac{s}{8 + 2s^2} \right) = \frac{10}{s}$$

$$V(s) \left[\frac{8 + 2s^2 + 8s}{8(8 + 2s^2)} \right] = \frac{10}{s}$$

Simplify further,

$$V(s) \left[\frac{4 + s^2 + 4s}{4(8 + 2s^2)} \right] = \frac{10}{s}$$

$$V(s) = \frac{40(8 + 2s^2)}{s(s^2 + 4s + 4)}$$

$$= \frac{80(s^2 + 4)}{s(s + 2)^2}$$

$$= \frac{80}{s} - \frac{320}{(s + 2)^2}$$

Apply the inverse Laplace transform.

$$\begin{aligned} v(t) &= 80u(t) - 320e^{-2t}tu(t) \\ &= 80(1 - 4te^{-2t})u(t) \text{ V} \end{aligned}$$

Therefore, the voltage across the resistor is $\boxed{80(1 - 4te^{-2t})u(t) \text{ V}}$.